



UNIVERSIDAD TORCUATO DI TELLA  
ESCUELA DE NEGOCIOS

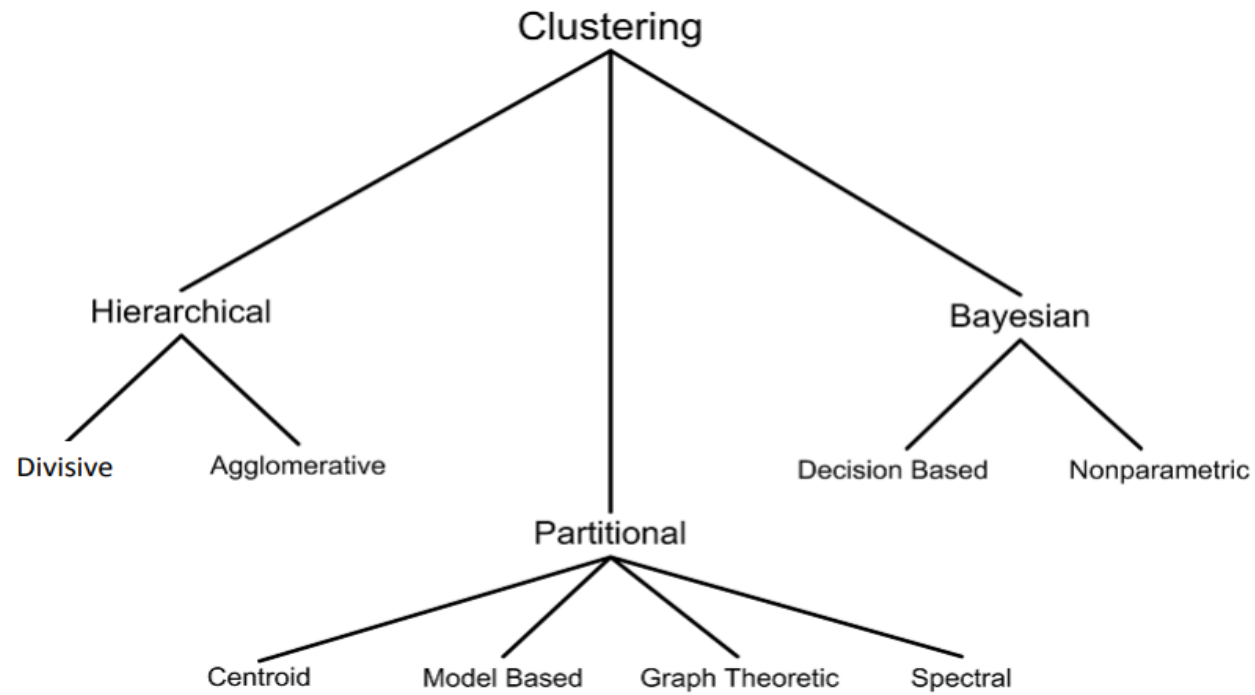
# **Intro a ciencia de datos**

**Prof. Pablo Roccatagliata**

**Maestría en Finanzas**

# CONTENIDOS PARTE IV CONTINUACIÓN

- Clustering



# DETECCIÓN DE FRAUDE

¿Tengo identificados todos los fraudes anteriores?

¿Y si no? ¿Còmo puedo usar un algoritmo para resolver el problema?

|                      |                  | <i>Forecast Result</i> |                  |
|----------------------|------------------|------------------------|------------------|
|                      |                  | <i>Fraud</i>           | <i>Non-Fraud</i> |
| <i>Practice Data</i> | <i>Fraud</i>     | TP                     | FN               |
|                      | <i>Non-Fraud</i> | FP                     | TN               |

# SEGMENTACIÓN DE CLIENTES

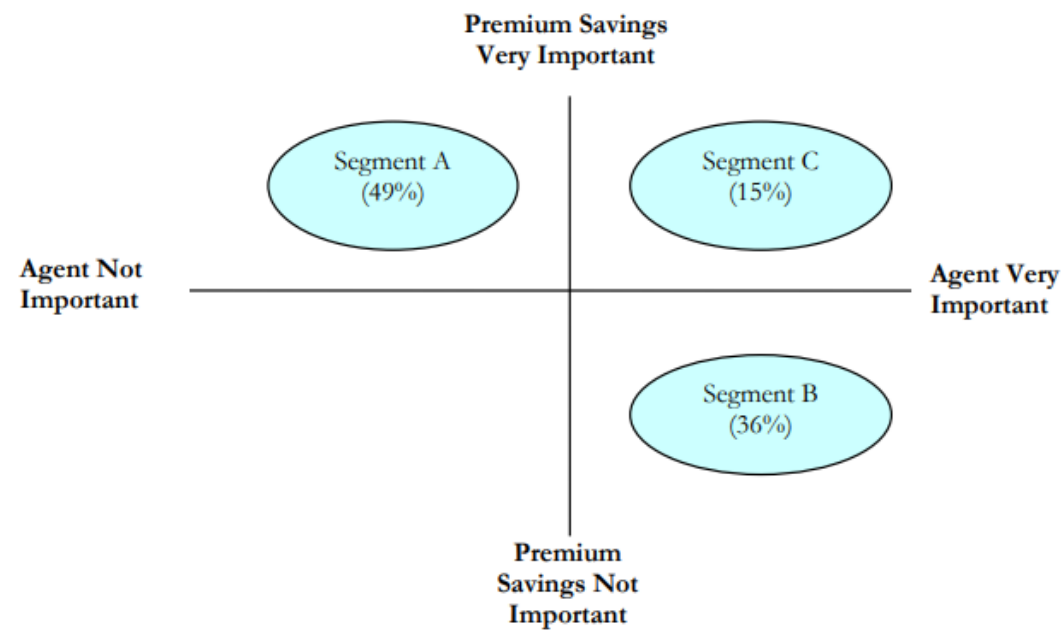


Table 1. Sample data for cluster analysis.

| Individual Name | Importance Score |                    |
|-----------------|------------------|--------------------|
|                 | Premium Savings  | Neighborhood Agent |
| Joe             | 4                | 7                  |
| Sam             | 3                | 4                  |
| Sara            | 5                | 3                  |

Table 3. Age and income distribution for segments.

| Segment | Mean |             | Range |                |
|---------|------|-------------|-------|----------------|
|         | Age  | Income (\$) | Age   | Income (\$)    |
| A       | 21   | 15,000      | 16–25 | 0–25,000       |
| B       | 45   | 120,000     | 33–55 | 75,000–215,000 |
| C       | 39   | 40,000      | 39–54 | 24,000–60,000  |

Fuente: Cluster Analysis for segmentation. Darden School of Business.

# MEJORAS EN LA GESTIÓN

## Tax Fraud Detection for Under-Reporting Declarations Using an Unsupervised Machine Learning Approach

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## CLUSTERING



- A través de las tarjetas de membresía tiene algunos datos básicos sobre sus clientes
- Se desea segmentar a los clientes para poder planificar estrategias de marketing

# Lo revolvemos!

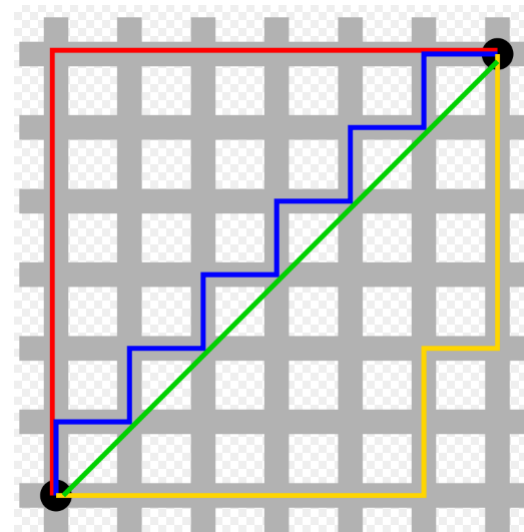
[Mall\\_costumers.csv](#)



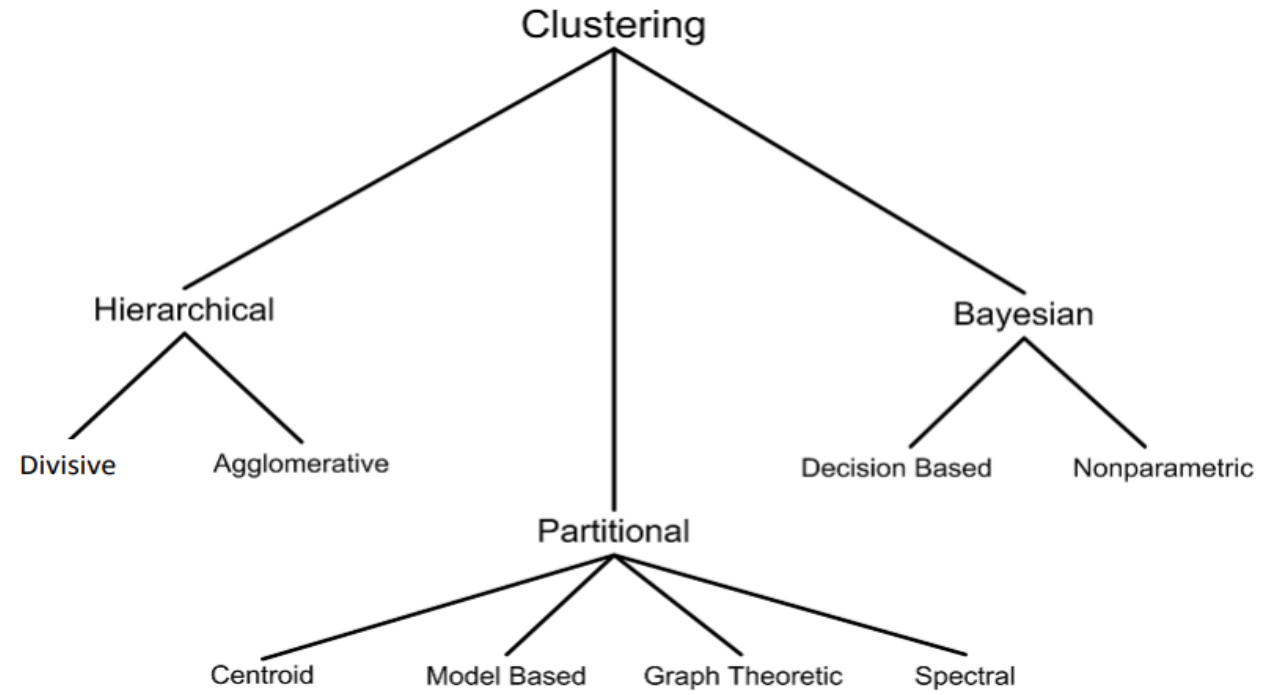
# CLUSTERING

- Queremos juntar parecidos
- Implicar definir medidas de distancia apropiadas para cada problema

|           |                                                     |
|-----------|-----------------------------------------------------|
| Euclidean | $\sqrt{\sum_{i=1}^k (x_i - y_i)^2}$                 |
| Manhattan | $\sum_{i=1}^k  x_i - y_i $                          |
| Minkowski | $\left( \sum_{i=1}^k ( x_i - y_i )^q \right)^{1/q}$ |

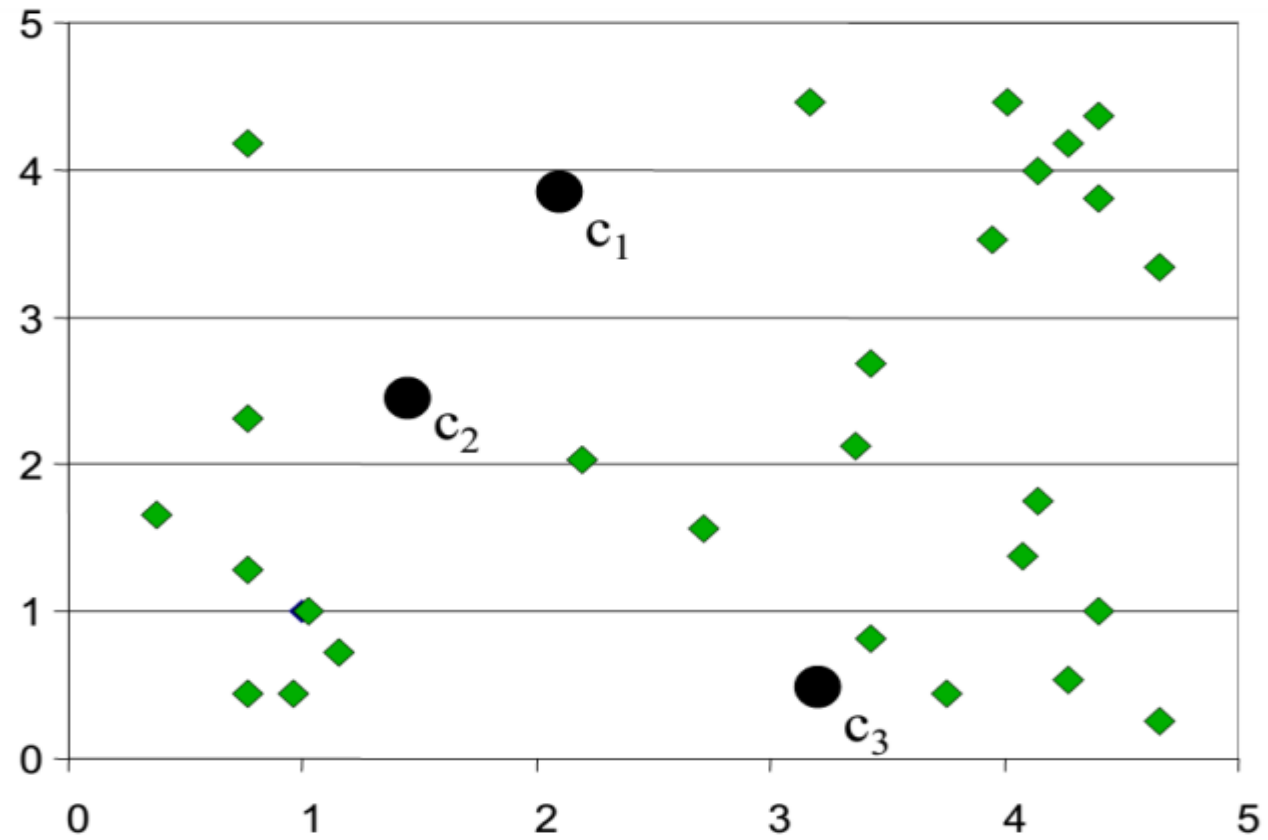


# CLUSTERING



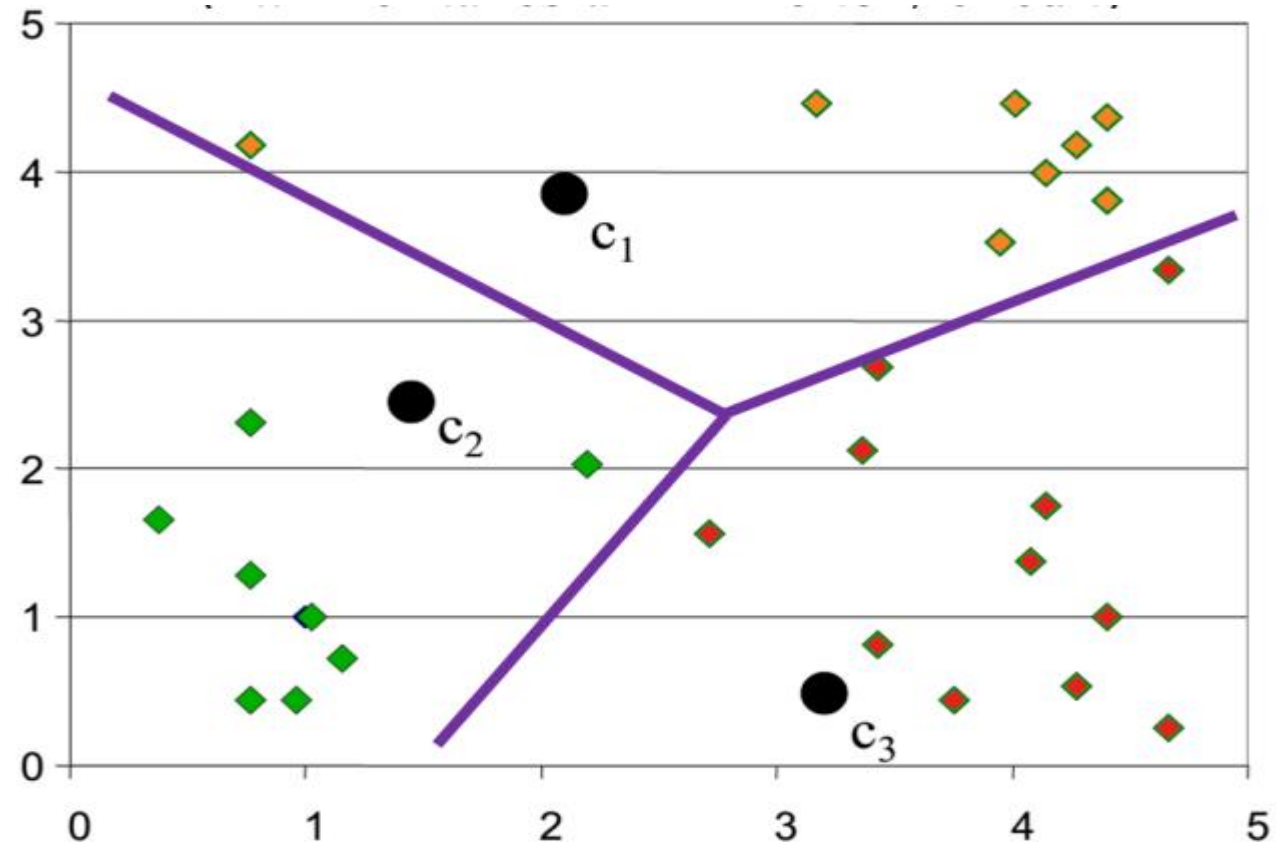
# CLUSTERING

## K-means



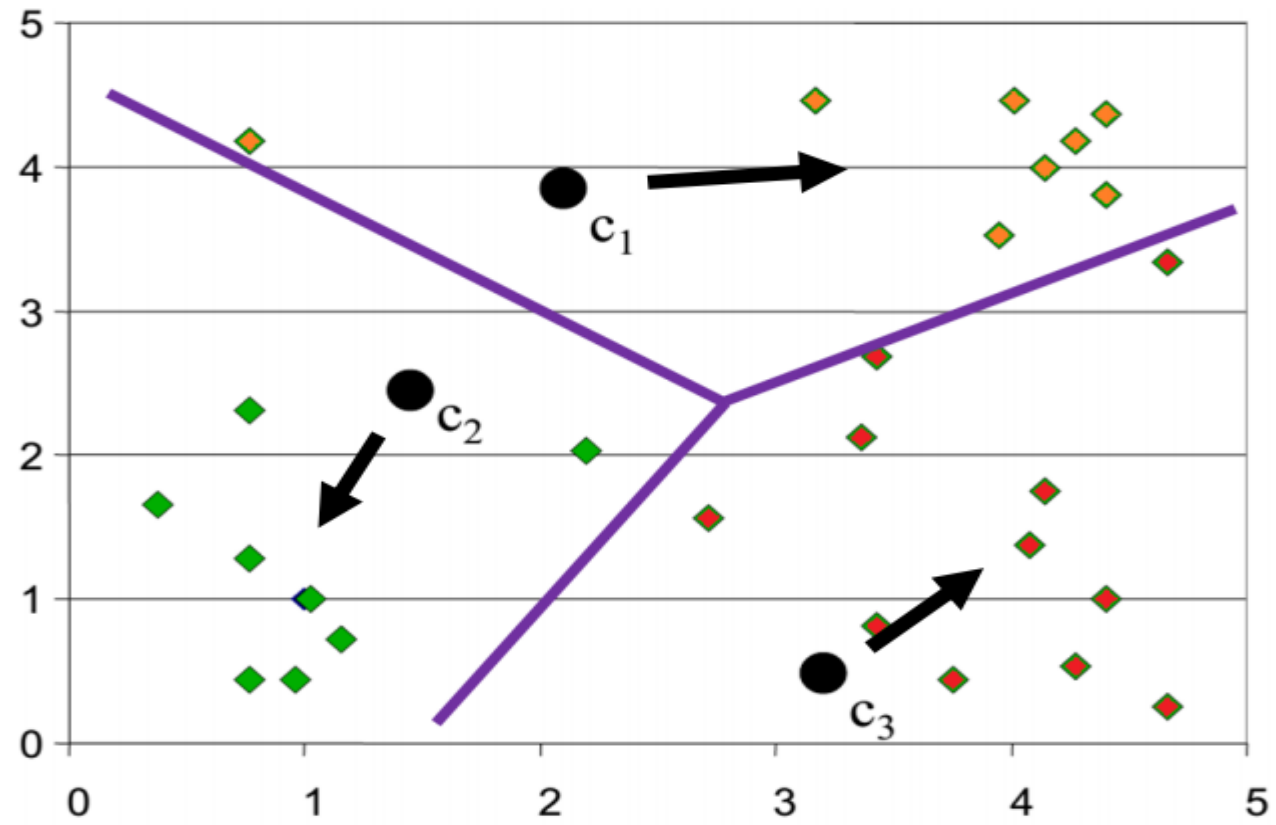
# CLUSTERING

## K-means



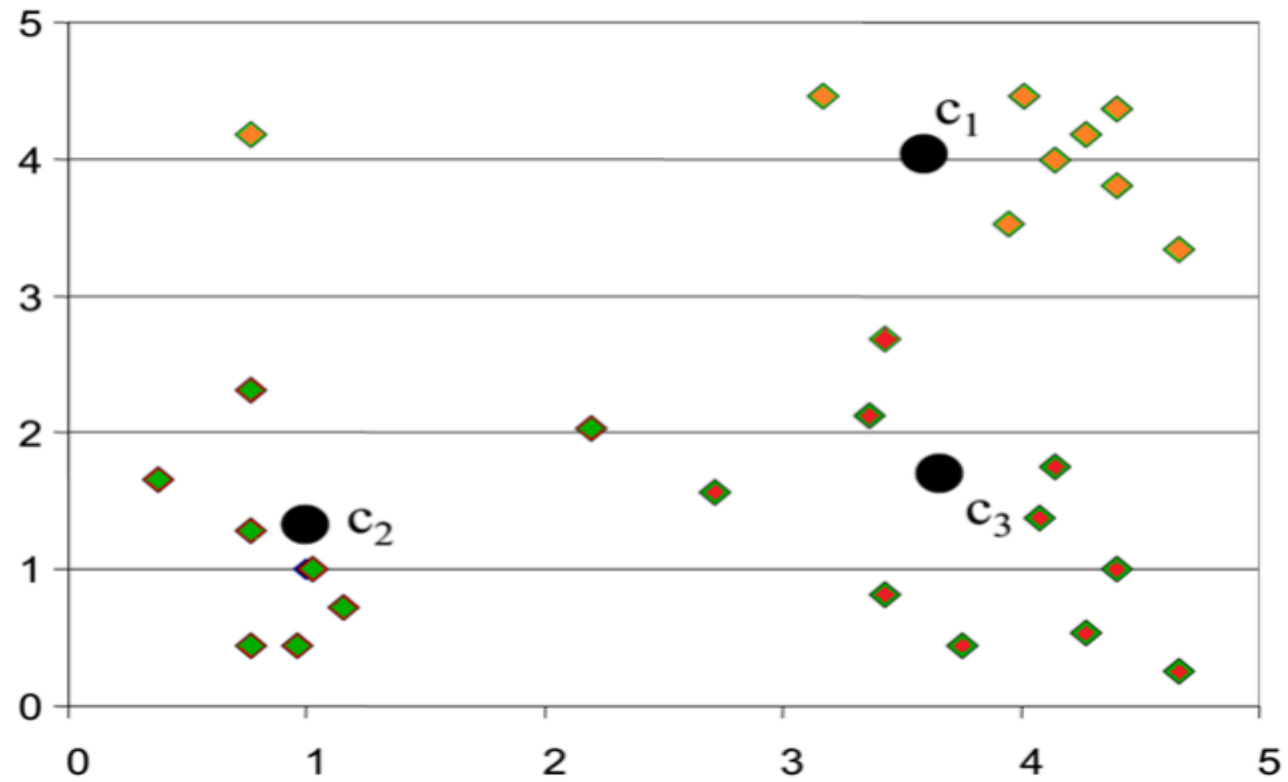
# CLUSTERING

## K-means



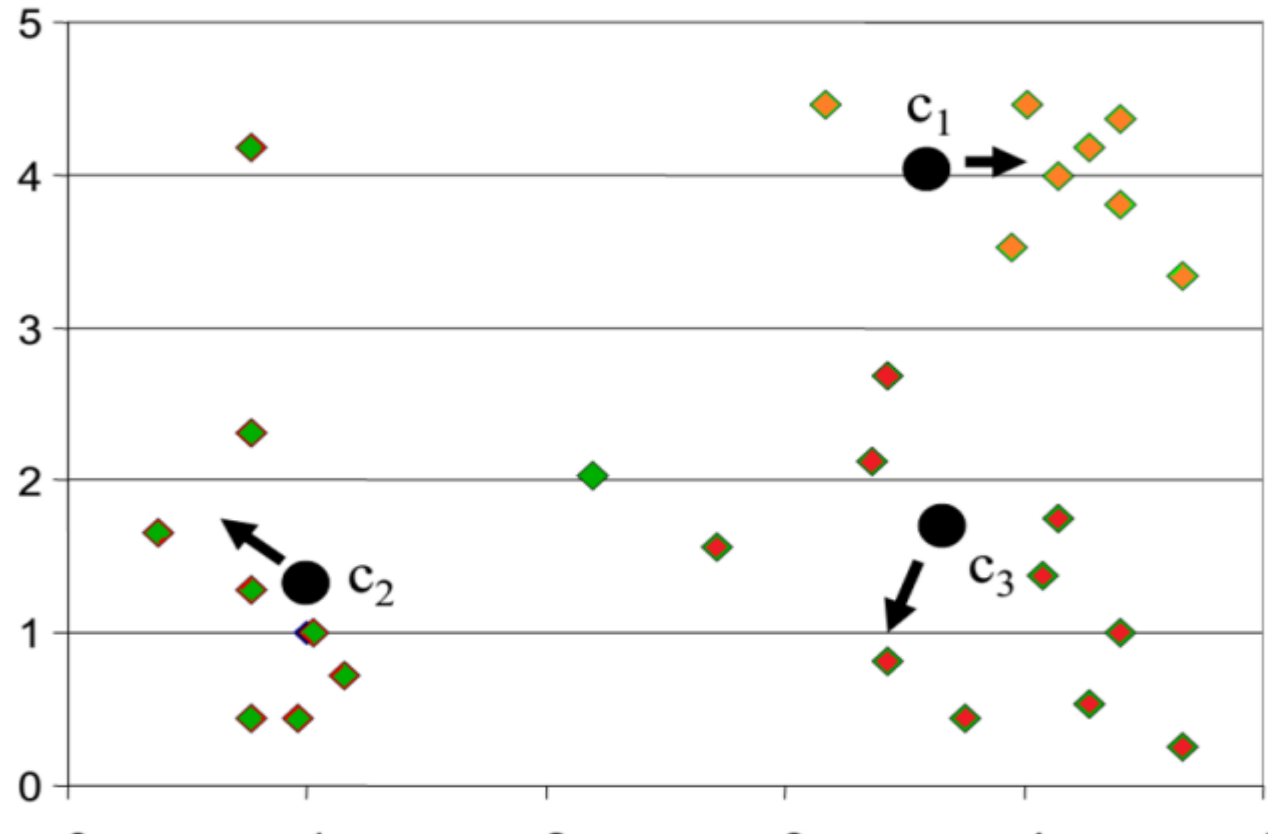
# CLUSTERING

## K-means



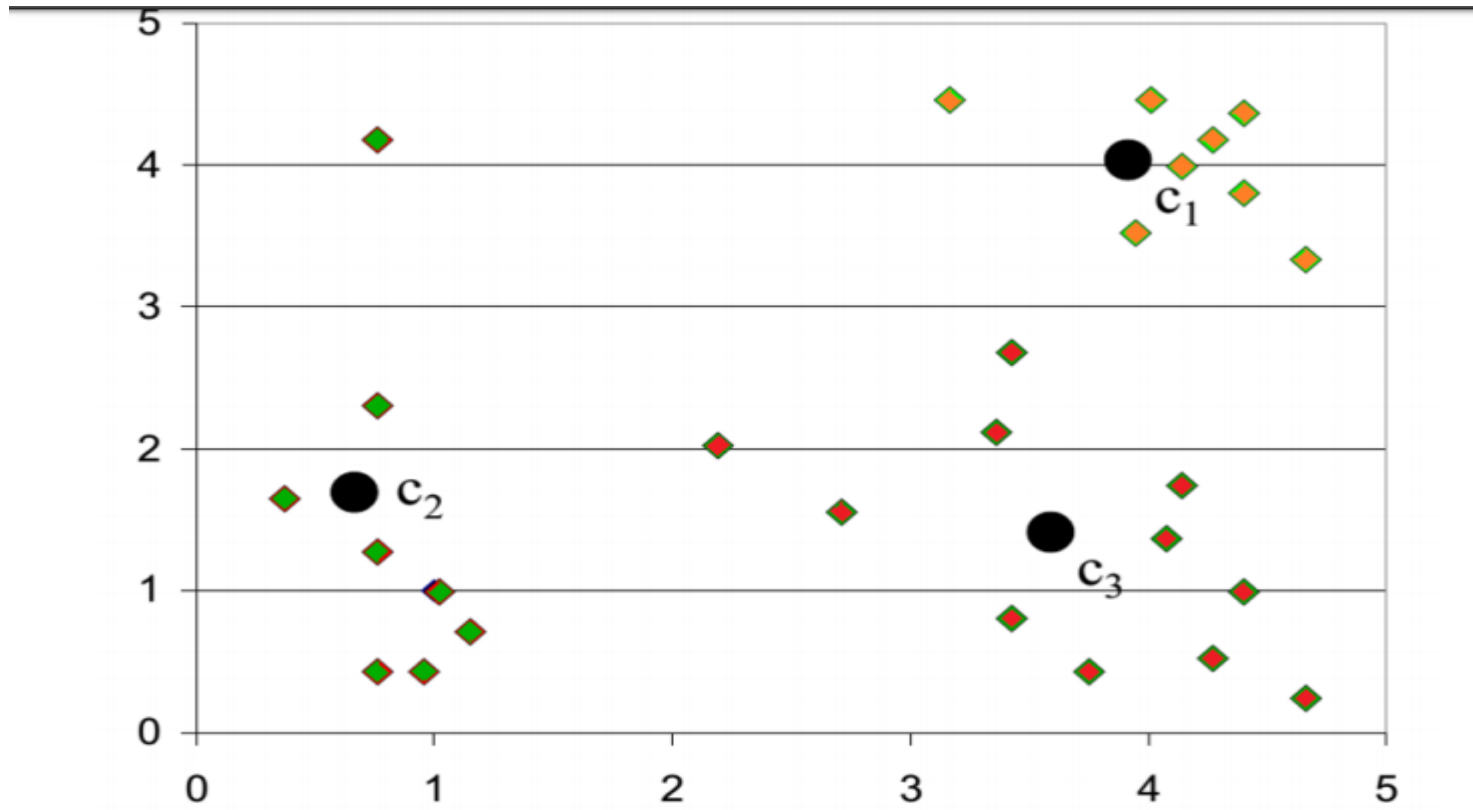
# CLUSTERING

## K-means



# CLUSTERING

## K-means



# CLUSTERING

## K-means

### Ventajas

- Simplicidad
- Eficiencia

### Desventajas

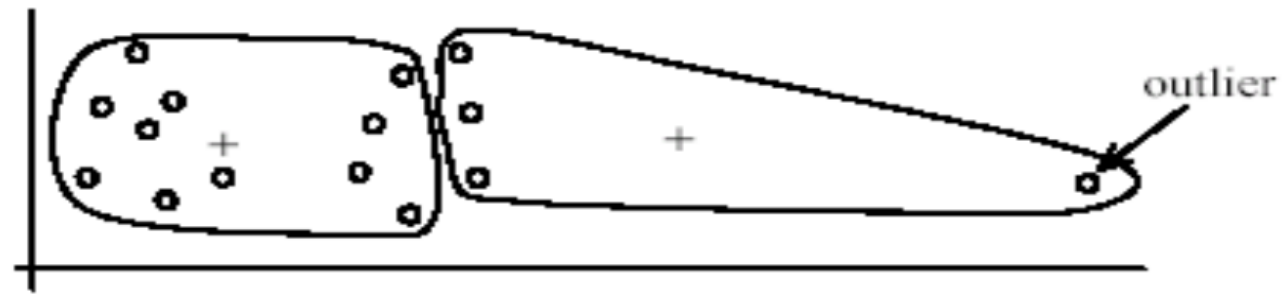
- Los resultados varían entre corridas
- Hay que elegir K
- Es sensible a valores atípicos
- El mínimo global es difícil de lograr

The diagram shows the objective function formula for K-means clustering:  $J = \sum_{j=1}^k \sum_{i=1}^n \|x_i^{(j)} - c_j\|^2$ . Annotations include: 'number of clusters' pointing to  $k$ , 'number of cases' pointing to  $n$ , 'case  $i$ ' pointing to  $x_i^{(j)}$ , 'centroid for cluster  $j$ ' pointing to  $c_j$ , 'objective function' pointing to  $J$ , and 'Distance function' pointing to the term  $\|x_i^{(j)} - c_j\|^2$ .

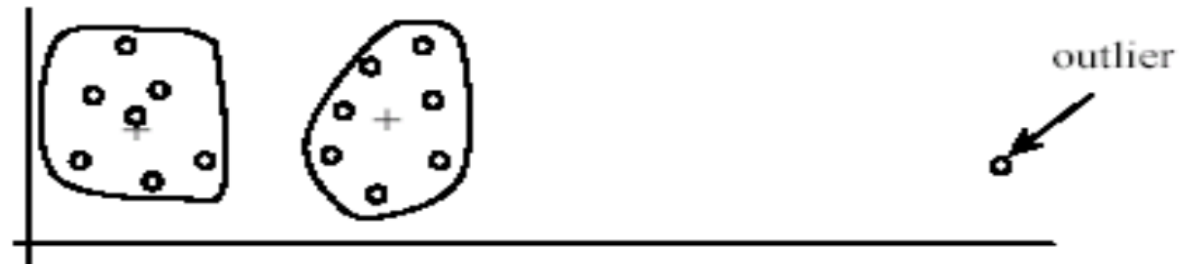
$$\text{objective function} \leftarrow J = \sum_{j=1}^k \sum_{i=1}^n \underbrace{\|x_i^{(j)} - c_j\|^2}_{\text{Distance function}}$$

# CLUSTERING

## K-means



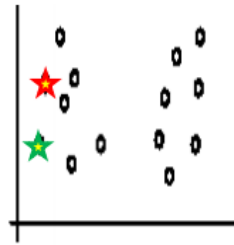
(A): Undesirable clusters



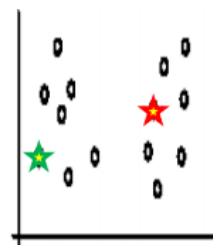
(B): Ideal clusters

# CLUSTERING

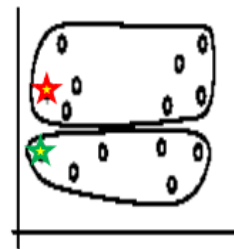
## K-means



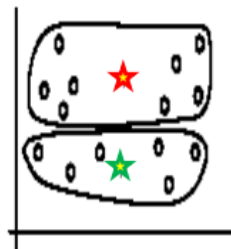
Random selection of seeds (centroids)



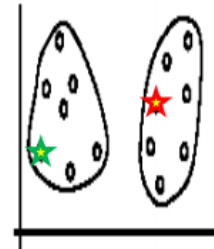
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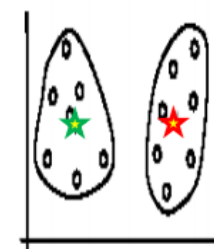
Iteration 1



Iteration 2



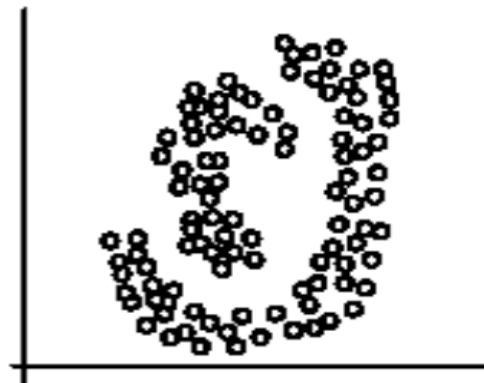
Iteration 1



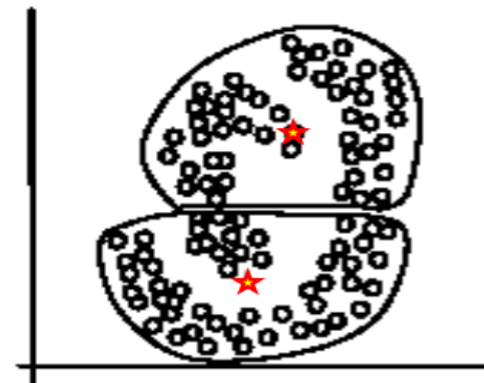
Iteration 2

# CLUSTERING

## K-means



(A): Two natural clusters



(B):  $k$ -means clusters

K-means tiene dificultades cuando los clusters no tienen formas esféricas

# CLUSTERING

## Evaluación de clusters

- Visualización
  - si tenemos pocas dimensiones
- Validación interna
  - Silhoutte score
  - Calinski Harabasz
- Validación externa

## CLUSTERING

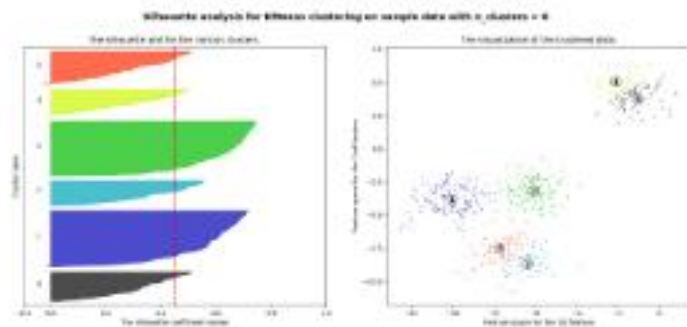
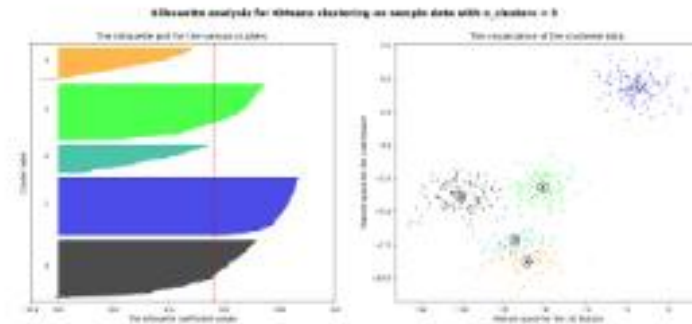
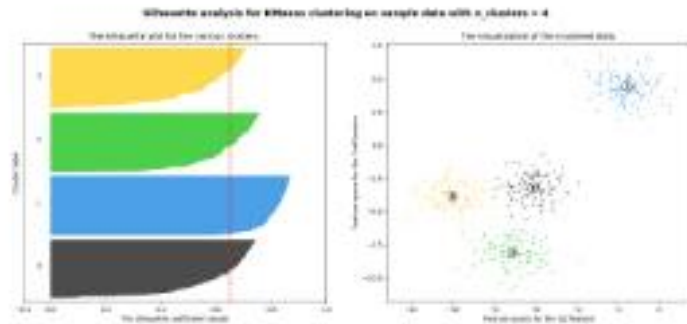
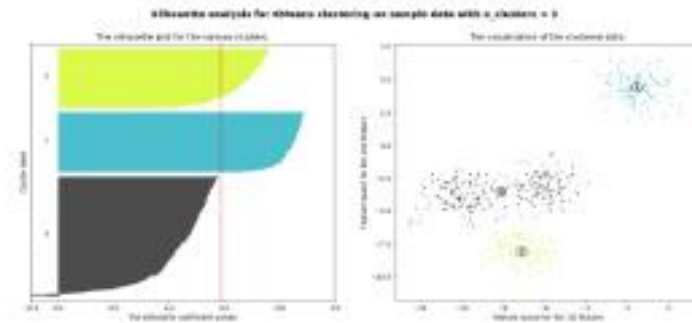
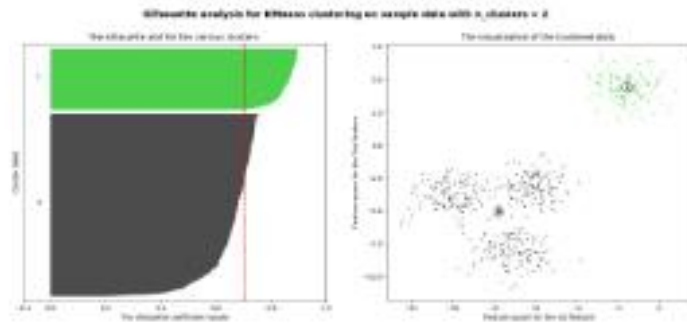
Silhouette score

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}},$$

$$s(i) = \begin{cases} 1 - a(i)/b(i), & \text{if } a(i) < b(i) \\ 0, & \text{if } a(i) = b(i) \\ b(i)/a(i) - 1, & \text{if } a(i) > b(i) \end{cases}$$

# CLUSTERING

## Silhoutte score



## CLUSTERING

Calinski Harabasz

$n$  = # data points  $k$  = # clusters

$W(k)$  = within cluster variation

$B(k)$  = between cluster variation

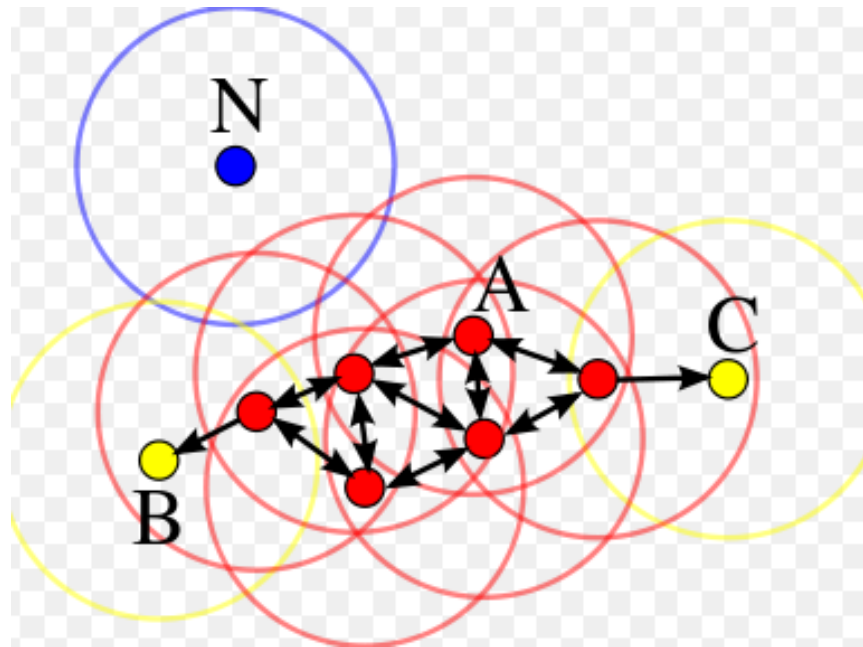
$$CH(k) = [B(k)/W(k)] \times [(n - k)/(k - 1)]$$

# CLUSTERING

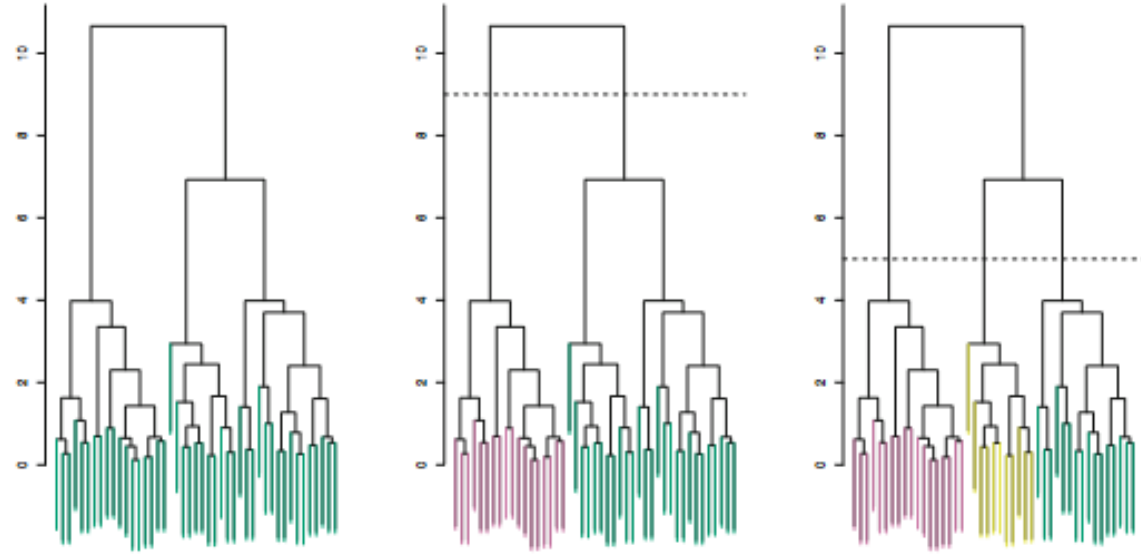
¿Si no sabemos cuántos clusters puede haber?

Podemos usar DBSCAN

<https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/>

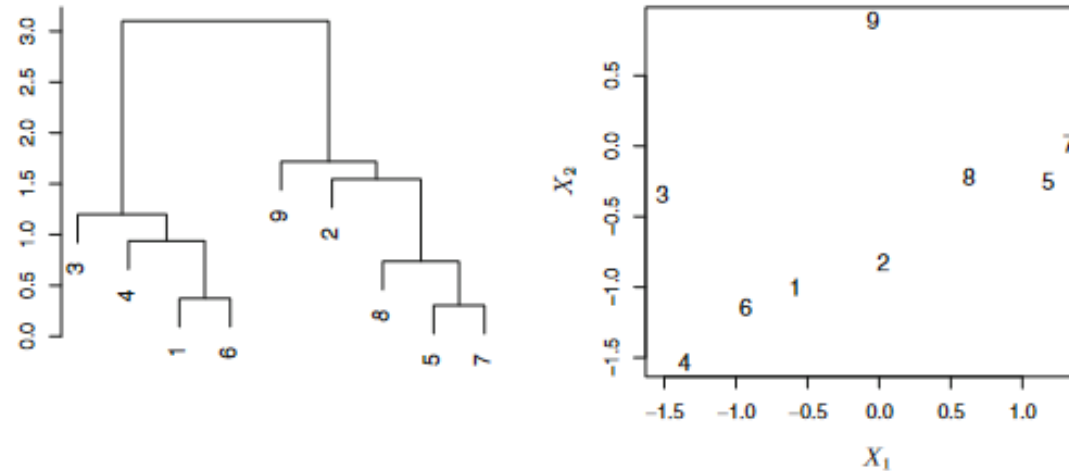


## CLUSTERING JERÁRQUICO



**FIGURE 10.9.** Left: dendrogram obtained from hierarchically clustering the data from Figure 10.8 with complete linkage and Euclidean distance. Center: the dendrogram from the left-hand panel, cut at a height of nine (indicated by the dashed line). This cut results in two distinct clusters, shown in different colors. Right: the dendrogram from the left-hand panel, now cut at a height of five. This cut results in three distinct clusters, shown in different colors. Note that the colors were not used in clustering, but are simply used for display purposes in this figure.

# CLUSTERING JERÁRQUICO



**FIGURE 10.10.** An illustration of how to properly interpret a dendrogram with nine observations in two-dimensional space. Left: a dendrogram generated using Euclidean distance and complete linkage. Observations 5 and 7 are quite similar to each other, as are observations 1 and 6. However, observation 9 is no more similar to observation 2 than it is to observations 8, 5, and 7, even though observations 9 and 2 are close together in terms of horizontal distance. This is because observations 2, 8, 5, and 7 all fuse with observation 9 at the same height, approximately 1.8. Right: the raw data used to generate the dendrogram can be used to confirm that indeed, observation 9 is no more similar to observation 2 than it is to observations 8, 5, and 7.

# CLUSTERING JERÁRQUICO

¿Distancias entre pares de grupos de observaciones?

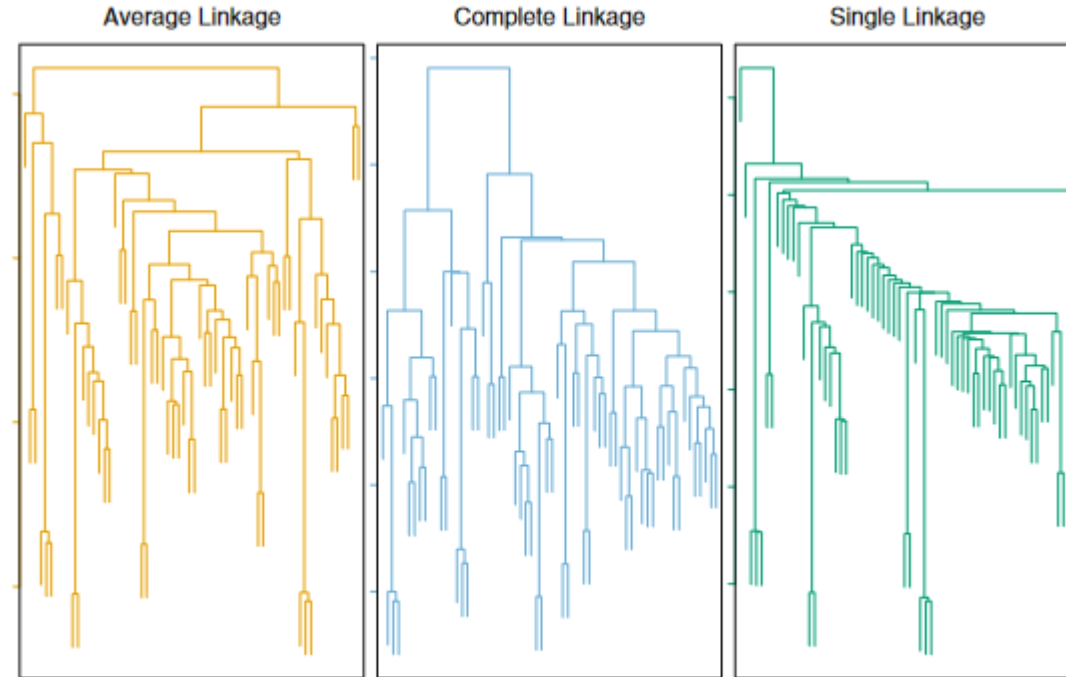
Linkage: dissimilarity between two groups of observations

| <i>Linkage</i> | <i>Description</i>                                                                                                                                                                                                                                                                                                        |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complete       | Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>largest</i> of these dissimilarities.                                                                                                                 |
| Single         | Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>smallest</i> of these dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one-at-a-time. |
| Average        | Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>average</i> of these dissimilarities.                                                                                                                    |
| Centroid       | Dissimilarity between the centroid for cluster A (a mean vector of length $p$ ) and the centroid for cluster B. Centroid linkage can result in undesirable <i>inversions</i> .                                                                                                                                            |

**TABLE 10.2.** A summary of the four most commonly-used types of linkage in hierarchical clustering.

# CLUSTERING JERÁRQUICO

Linkage: dissimilarity between two groups of observations



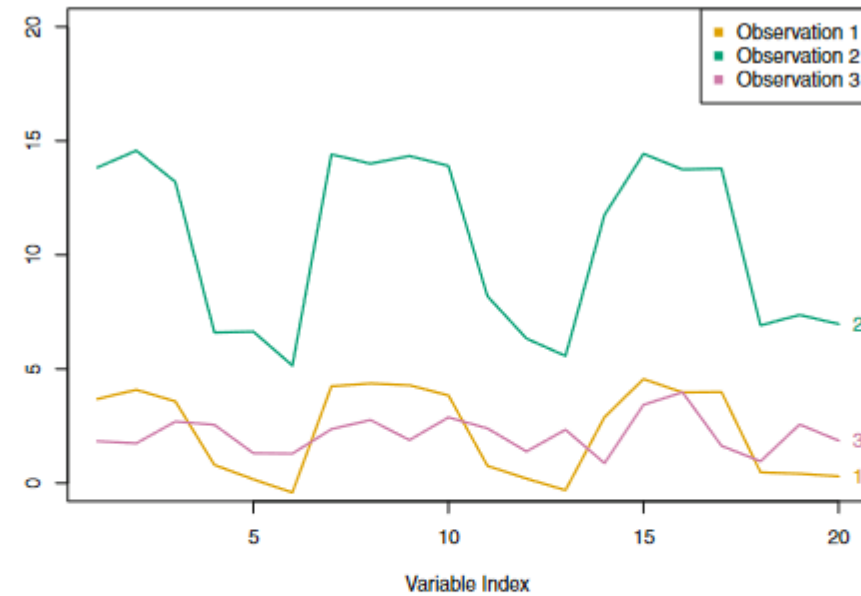
**FIGURE 10.12.** Average, complete, and single linkage applied to an example data set. Average and complete linkage tend to yield more balanced clusters.

## CHOICE OF DISSIMILARITY MEASURE

¿Alternativas a distancia euclidena como la dissimilarity measure?

**Correlation-based distance** considera que 2 observaciones son similares si sus features están correlacionados, aún sin los valores observados de los features están lejos en términos de la distancia euclidean

¿Necesitamos re escalar además?



**FIGURE 10.13.** Three observations with measurements on 20 variables are shown. Observations 1 and 3 have similar values for each variable and so there is a small Euclidean distance between them. But they are very weakly correlated, so they have a large correlation-based distance. On the other hand, observations 1 and 2 have quite different values for each variable, and so there is a large Euclidean distance between them. But they are highly correlated, so there is a small correlation-based distance between them.

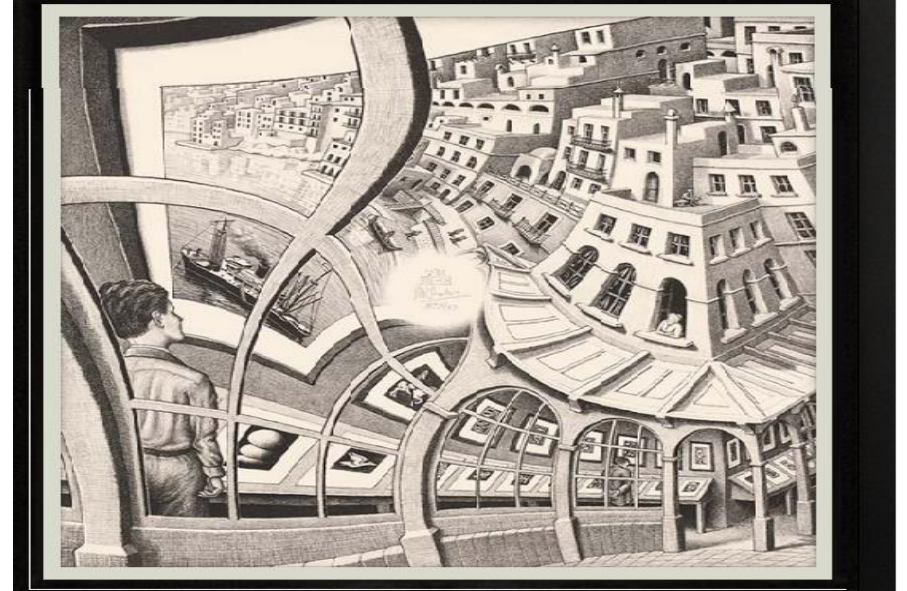
**¿Preguntas?**



# Gracias!



[proccatagliata@gmail.com](mailto:proccatagliata@gmail.com)



Gallery. M.C. Escher

# **Anexo**

## **Quiz problemas no supervisados**



## REPASO

*En PCA hace falta normalizar los features para que tengan la misma varianza*

- Falso
- Verdadero.
- Depende del tamaño muestral.
- No puede responderse con la información disponible.

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*Si usamos PCA para generar los predictores entonces debemos hacer el Split entre train y test antes de generar los componentes principales.*

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## REPASO

*¿Cuál de los siguientes es un ejemplo de un algoritmo determinístico?*

- PCA
- K-means
- K-vecinos
- Ninguna de las anteriores

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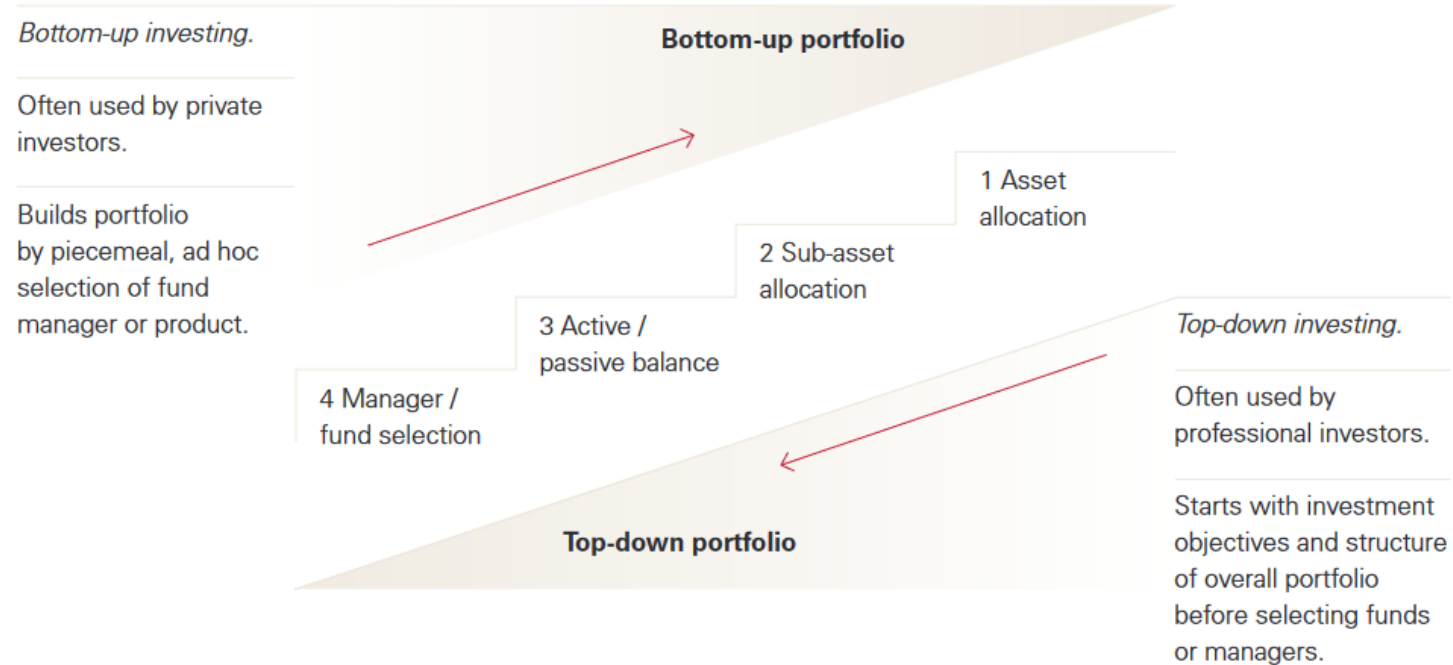
# **Anexo**

## **Construcción de portafolios: Hierarchical risk parity (HRP)**

**Veremos el detalle  
en Financial  
Analytics!**



# ASSET ALLOCATION



¿Dónde entra lo que vamos a ver hoy?

Off topic: ¿ qué tan importante es asset allocation empíricamente?

¿90%? ¿50%?

[Respuesta Xiong \(2010\).](#)

[Fuente: Portfolio construction A systematic approach to investing. Vanguard.](#)

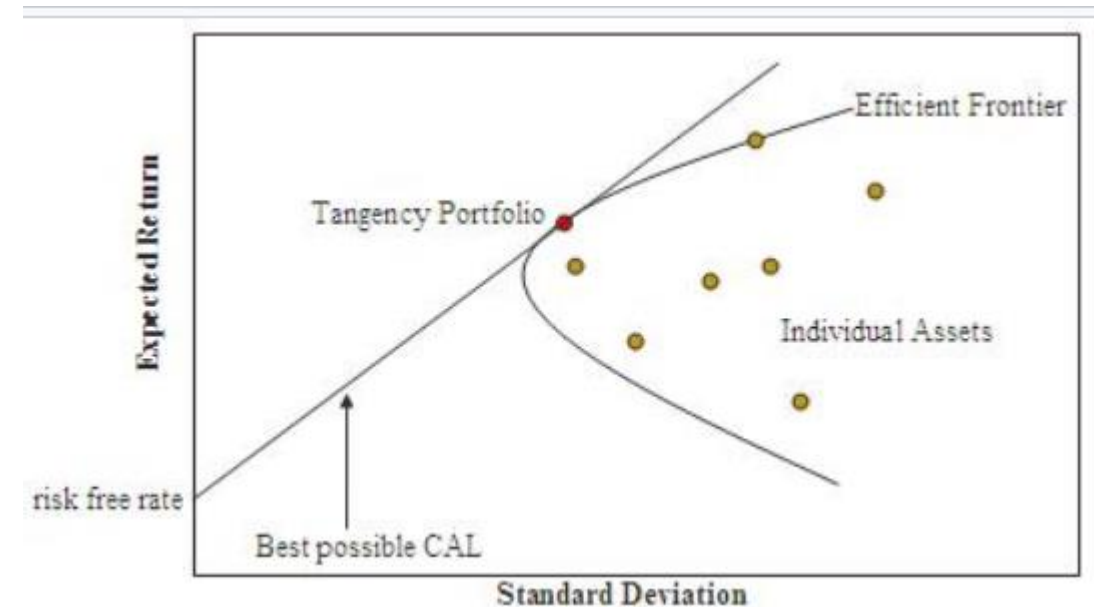
## ASSET ALLOCATION

- La construcción de carteras es quizás el problema financiero más recurrente. Diariamente, los administradores de inversiones deben construir carteras que incorporen sus puntos de vista y pronósticos sobre riesgos y retornos.
- Principales estrategias de construcción de portafolios
  - Modern Portfolio Theory
  - Black and Litterman model.
  - Simple return-agnostic strategies.
  - Strategies based on higher order moments
  - Factor-based strategies.
  - **Risk-based strategies**
  - Growth Optimal Portfolio Theory



## ASSET ALLOCATION

- Principales estrategias de construcción de portafolios
  - Modern Portfolio Theory (Markowitz 1952)
    - ¿Qué problema enfrentamos aquí?
      - Minimizar varianza sujeto a un objetivo de retornos
      - ¿Alguna vez se pregunto cómo se resuelve el problema?
      - Markowitz's Critical Line Algorithm (CLA) que asegura la convergencia de los weights luego de una cantidad finita de pasos.
      - Problemas de los optimizadores cuadráticos y del CLA en particular:
        - **instability, concentration, and under-performance.**

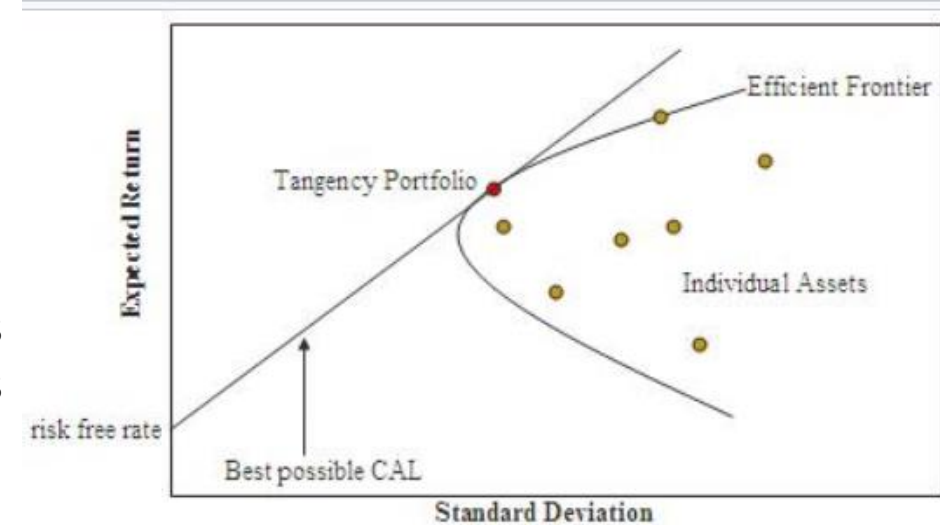


Avance: importan las correlaciones y no solamente los retornos esperados!

Introduce

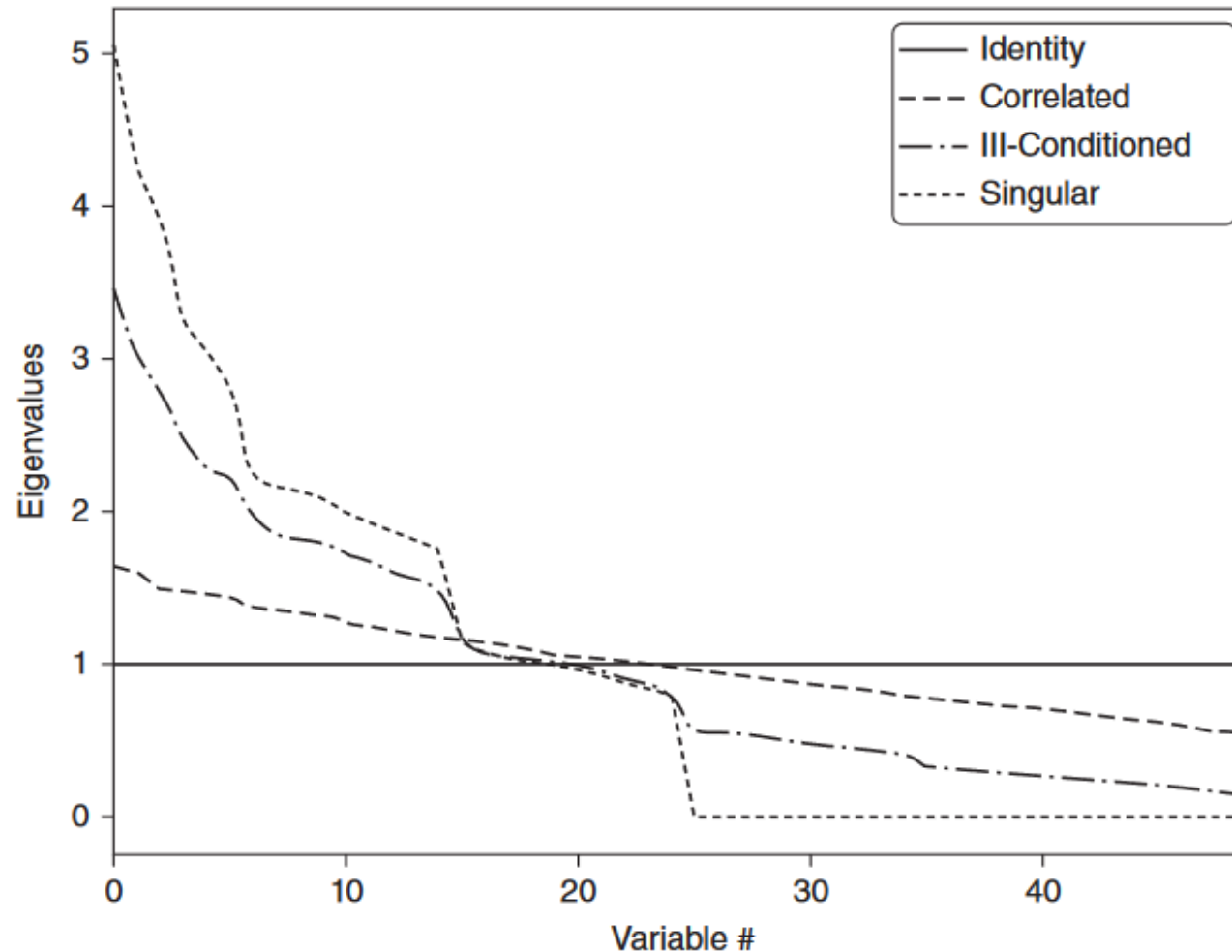
# ASSET ALLOCATION

- Principales estrategias de construcción de portafolios
  - Modern Portfolio Theory (Markowitz 1952)
    - ¿Qué problema enfrentamos aquí?
      - Problemas de los optimizadores cuadráticos y del CLA en particular:
        - Instability: pequeños desvios de los retornos respect a los forecasts tienen un gran impacto en los portafolios generados por el CLA.
        - De ahí la idea de concentrarnos en la matriz de covarianzas y olvidarnos de los forecasts de retornos!
        - Enfoques como risk parity entonces mejoran el problema de inestabilidad pero no lo resuelven!
        - En métodos cuadráticos seguimos teniendo que invertir una matriz de covarianzas definida positiva, esta inversion es inestable cuando la matriz tiene condition number alto



Avance: importan las correlaciones y no solamente los retornos esperados!

## ASSET ALLOCATION



**FIGURE 16.1** Visualization of Markowitz's curse

A diagonal correlation matrix has the lowest condition number. As we add correlated investments, the maximum eigenvalue is greater and the minimum eigenvalue is lower. The condition number rises quickly, leading to unstable inverse correlation matrices. At some point, the benefits of diversification are more than offset by estimation errors.

¿Por qué se habla de Markowitz's Curse?

A medida que se agregan más activos correlacionados, sube la necesidad de diversificación pero como aumenta el [condition number](#), es más probable recibir soluciones inestables

Los beneficios de la diversificación son más que compensados por los errores de estimación.

Fuente: Marcos López de Prado  
Advances in Financial Machine Learning

# ASSET ALLOCATION

## Principales estrategias de construcción de portafolios

- Modern Portfolio Theory (Markowitz 1952)
- ¿Qué problema enfrentamos aquí?
  - Instability. Aumentar el tamaño de la matriz de covarianza solo empeorará las cosas, ya que cada coeficiente de covarianza se estima con menos grados de libertad.
  - En general, necesitamos al menos  $\frac{1}{2}N(N + 1)$  observaciones independientes e idénticamente distribuidas (IID) para estimar una matriz de covarianza de tamaño N que no es singular.
  - Por ejemplo, la estimación de una matriz de covarianza invertible de tamaño 50 requiere, como mínimo, 5 años de datos IID diarios.



|       |      |         |       |      |
|-------|------|---------|-------|------|
| 51.36 | 1.36 | +180.98 | -0.21 | 4.75 |
| 21.88 | 5.56 | +740.21 | -6.87 | 8.87 |
| 78.69 | 8.24 | +122.56 | -9.45 | 1.54 |
| 18.75 | 9.62 | +140.04 | -3.36 | 7.02 |
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# ASSET ALLOCATION

## Principales estrategias de construcción de portafolios

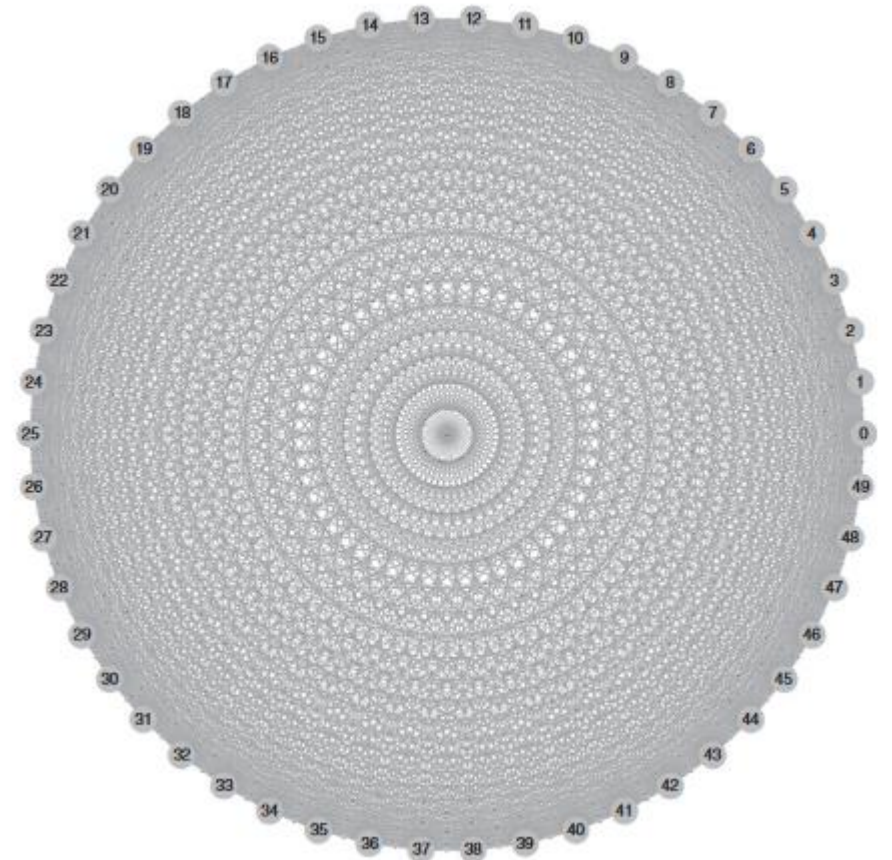
- Modern Portfolio Theory (Markowitz 1952)
- ¿Qué problema enfrentamos aquí?
  - Como saben la mayoría de los inversores, las estructuras de correlación no permanecen invariables durante períodos tan largos con un nivel de confianza razonable.
  - incluso las carteras ingenuas (igualmente ponderadas) superan la varianza media y la optimización basada en el riesgo fuera de la muestra (De Miguelet al. [2009]).

## ASSET ALLOCATION

- Modern Portfolio Theory (Markowitz 1952)
- Todo empezaba con las series de precios...
- Luego los retornos...
- Luego la matriz de covarianzas de los retornos!
- ¿Qué representa esa matriz?

*A correlation matrix is a linear algebra object that measures the cosines of the angles between any two vectors in the vector space formed by the returns series ¿y si dropeamos edges innecesarios?*

The complete-graph- Correlation matrices can be represented as complete graphs, which lack the notion of hierarchy: Each investment is substitutable with another. Relationships implied by a covariance matrix of  $50 \times 50$ , that is 50 nodes and 1225 edges.



Fuente: Marcos López de Prado  
Advances in Financial Machine  
Learning

## ¿SON IGUALMENTE SUSTITUTOS ENTRE SÍ LOS N ACTIVOS?

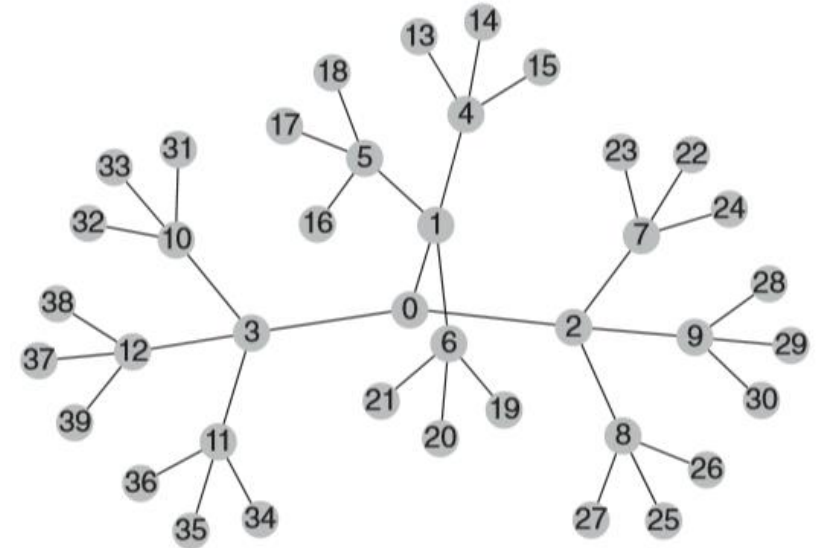
- Suponga que un inversor desea construir una cartera diversificada de valores, que incluya cientos de acciones, bonos, fondos de cobertura, bienes raíces, colocaciones privadas, etc.
- Algunas inversiones parecen sustitutos más cercanos entre sí, y otras inversiones parecen complementarias entre sí.
- Por ejemplo, las acciones podrían agruparse en términos de liquidez, tamaño, industria y región, donde las acciones dentro de un grupo determinado compiten por la asignación.



[Fuente imagen](#)

## ¿SON IGUALMENTE SUSTITUTOS ENTRE SÍ LOS N ACTIVOS?

- Ventajas de la estructura de árbol
  - Sólo tiene  $N-1$  bordes para conectar  $N$  nodos, por lo que los weights solo se se rebalancean entre pares en varios niveles jerárquicos
  - Los weights se distribuyen de forma top down consistente con la forma en que muchos PM Construyen sus portafolios
  - En cambio en el grafo completo los weights pueden variar libremente lo que índice la inestabilidad de los resultados del CLA



Fuente: Advances in Financial Machine Learning.  
Marcos López de Prado.

## HIERARCHICAL RISK PARITY

- Pertence al risk-based asset allocation approaches
- Objetivo: diversified portfolio basado en la información contenida en la matriz de covarianzas.
- HRP no requiere la invertibilidad de la matriz de covarianzas de los retornos
- Los experimentos de Monte Carlo muestran que HRP ofrece una menor varianza out of sample que CLA, aunque la varianza mínima es el objetivo de optimización de CLA.
- HRP produce portafolios de menor riesgo en comparación con los métodos
- **Una aplicación conveniente de HRP es determinar asignaciones a través de múltiples estrategias de aprendizaje automático (ML)**



## HIERARCHICAL RISK PARITY

- Puede computar un portafolio incluso si la matriz de covarianzas es singular (no tiene inversa)
- Tomemos una matriz de dimensiones  $T \times N$  , por ejemplo los retornos de  $N$  activos durante  $T$  períodos y luego armemos la matrix de correlaciones con dimensiones  $N \times N$
- 3 pasos
  - tree clustering
  - quasi-diagonalization
  - recursive bisection.



## HIERARCHICAL RISK PARITY

Tree clustering Paso 1 ...

Construir matriz de distancias a partir de matriz de correlaciones de NxN con entradas  $\rho = \{\rho_{i,j}\}_{i,j=1,\dots,N}$

donde  $\rho_{i,j} = \rho[X_i, X_j]$ .

Usamos una metrica de distancia  $d : (X_i, X_j) \in B \rightarrow \mathbb{R} \in [0, 1], d_{i,j} = d[X_i, X_j] = \sqrt{\frac{1}{2}(1 - \rho_{i,j})}$ ,

Donde B es el producto cartesiano de los ítems en  $\{1, \dots, i, \dots, N\}$

Nota: Ojo que d tiene que cumplir propiedades de las métricas de distancia. Ver anexo del libro!

## HIERARCHICAL RISK PARITY

Tree clustering Paso 1 ...

Esta métrica de distancias nos permite construir una matriz nueva de distancias de NxN

$$D = \{d_{ij}\}_{i,j=1,\dots,N}$$

$$\{\rho_{ij}\} = \begin{bmatrix} 1 & .7 & .2 \\ .7 & 1 & -.2 \\ .2 & -.2 & 1 \end{bmatrix} \rightarrow \{d_{ij}\} = \begin{bmatrix} 0 & .3873 & .6325 \\ .3873 & 0 & .7746 \\ .6325 & .7746 & 0 \end{bmatrix}$$

## HIERARCHICAL RISK PARITY

Tree clustering Paso 2 ...

Distancias euclidianas entre las columnas de la matriz D (una distancia de distancias!)

$$\tilde{d} : (D_i, D_j) \subset B \rightarrow \mathbb{R} \in [0, \sqrt{N}], \tilde{d}_{i,j} = \tilde{d}[D_i, D_j] = \sqrt{\sum_{n=1}^N (d_{n,i} - d_{n,j})^2}.$$

$$\{d_{i,j}\} = \begin{bmatrix} 0 & .3873 & .6325 \\ .3873 & 0 & .7746 \\ .6325 & .7746 & 0 \end{bmatrix} \rightarrow \{\tilde{d}_{i,j}\}_{i,j=\{1,2,3\}} = \begin{bmatrix} 0 & .5659 & .9747 \\ .5659 & 0 & 1.1225 \\ .9747 & 1.1225 & 0 \end{bmatrix}$$

Los  $d_{ij}$  ahora son función de toda la matriz de correlaciones y no de un solo de una pareja particular de correlaciones!

## HIERARCHICAL RISK PARITY

Tree clustering Paso 3 ...

Armamos clusteres en base a estas distancias! Las juntamos en pares  $(i^*, j^*)$  tales que

$$(i^*, j^*) = \operatorname{argmin}_{i \neq j} \{\tilde{d}_{i,j}\}$$

$$\{\tilde{d}_{i,j}\}_{i,j=\{1,2,3\}} = \begin{bmatrix} 0 & .5659 & .9747 \\ .5659 & 0 & 1.1225 \\ .9747 & 1.1225 & 0 \end{bmatrix} \rightarrow u[1] = (1, 2)$$

# HIERARCHICAL RISK PARITY

## Tree clustering Paso 4 ...

Definimos la distancia entre el nuevo cluster formado  $u[1]$  y los items aún no clusterizados de forma de que  $\{\tilde{d}_{i,j}\}$  pueda ser actualizada.

En el context de hierarchical clustering analysis, esto se llama “linkage criterion.”

Por ejemplo, podemos definir la distancia entre un item  $i$  y el nuevo cluster  $u[1]$  como  $\dot{d}_{i,u[1]} = \min[\{\tilde{d}_{i,j}\}_{j \in u[1]}]$

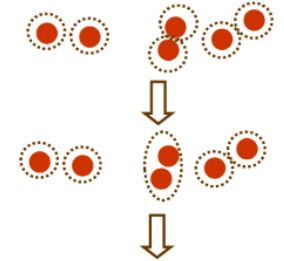
$$u[1] = (1, 2) \rightarrow \{\dot{d}_{i,u[1]}\} = \begin{bmatrix} \min [0, .5659] \\ \min [.5659, 0] \\ \min [.9747, 1.1225] \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ .9747 \end{bmatrix}$$

## Agglomerative hierarchical clustering

initialize with each example in singleton cluster

**while** there is more than 1 cluster

1. find 2 nearest clusters
2. merge them



### Four common ways to measure cluster distance

1. minimum distance  $d_{\min}(D_i, D_j) = \min_{x \in D_i, y \in D_j} \|x - y\|$
2. maximum distance  $d_{\max}(D_i, D_j) = \max_{x \in D_i, y \in D_j} \|x - y\|$
3. average distance  $d_{\text{avg}}(D_i, D_j) = \frac{1}{n_i n_j} \sum_{x \in D_i} \sum_{y \in D_j} \|x - y\|$
4. mean distance  $d_{\text{mean}}(D_i, D_j) = \|\mu_i - \mu_j\|$

Fuente: Lecture notes Unsupervised learnin  
Shimon Ullman + Tomaso Poggio Danny  
Harari + DaneilZysman + Darren Seiber

## HIERARCHICAL RISK PARITY

Tree clustering Paso 5 ...

Actualizamos la matriz  $\{\tilde{d}_{i,j}\}$  apendeando  $\dot{d}_{i,u[1]}$  y droppeando las filas y columnas clusterizadas  $j \in u[1]$

$$\{\tilde{d}_{i,j}\}_{i,j=\{1,2,3,4\}} = \begin{bmatrix} 0 & .5659 & .9747 & 0 \\ .5659 & 0 & 1.1225 & 0 \\ .9747 & 1.1225 & 0 & .9747 \\ 0 & 0 & .9747 & 0 \end{bmatrix}$$

$$\{\tilde{d}_{i,j}\}_{i,j=\{3,4\}} = \begin{bmatrix} 0 & .9747 \\ .9747 & 0 \end{bmatrix}$$

## HIERARCHICAL RISK PARITY

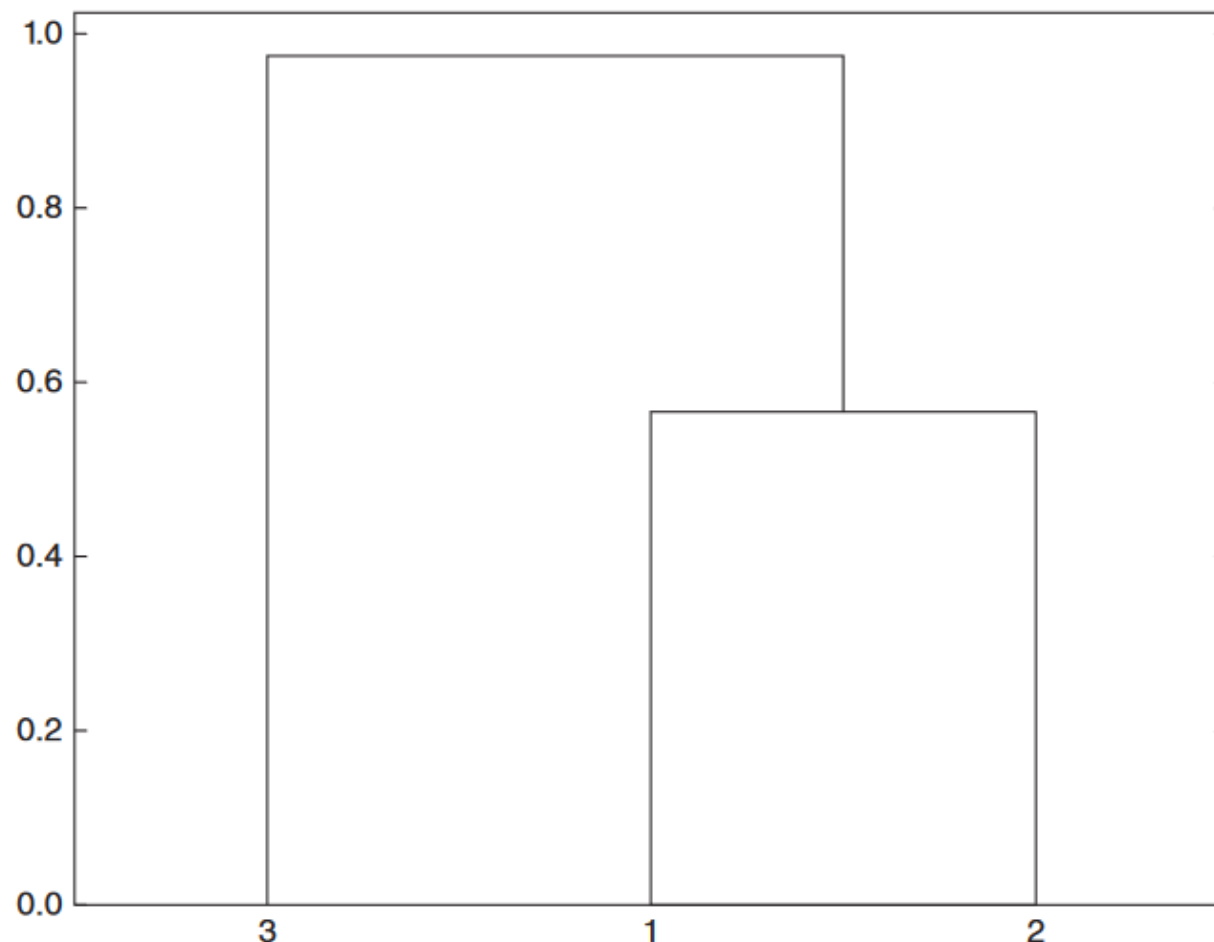
Tree clustering Paso 6 ...aplicamos recursivamente pasos 3,4 y 5 de nuevo..

Eventualmente todo forma parte de un mismo cluster

$$\{\tilde{d}_{ij}\}_{i,j=\{3,4\}} = \begin{bmatrix} 0 & .9747 \\ .9747 & 0 \end{bmatrix} \rightarrow u[2] = (3, 4) \rightarrow \text{Stop}$$

Ahora ordenamos los activos según sus parecidos y dividimos los weights recursivamente!

## HIERARCHICAL RISK PARITY



**FIGURE 16.3** Sequence of cluster formation

A tree structure derived from our numerical example, here plotted as a dendrogram. The y-axis measures the distance between the two merging leaves.

El output de la clusterizacion es una matriz de linkeage de  $(N-1) \times 4$  (4 tuple per cluster)

$$Y = \{(y_{m,1}, y_{m,2}, y_{m,3}, y_{m,4})\}_{m=1, \dots, N-1}$$

Los items  $(y_{m,1}, y_{m,2})$  indicant los integrantes de cada cluster

$y_{m,3} = \tilde{d}_{y_{m,1}, y_{m,2}}$  es la distancia

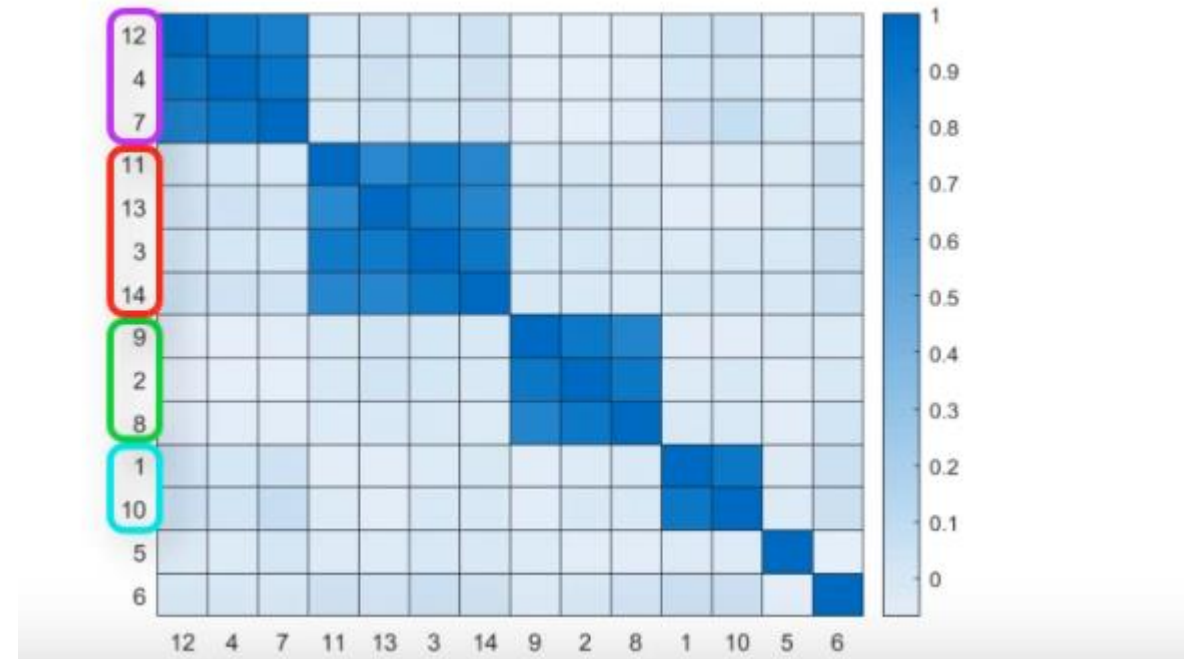
Item  $y_{m,4} \leq N$  indica los items originales incluidos en el cluster m

# HIERARCHICAL RISK PARITY

3 pasos

- **quasi-diagonalization**

- This stage reorganizes the rows and columns of the covariance matrix, so that the largest values lie along the diagonal.
- Similar invest-ments are placed together, and dissimilar investments are placed far apart.
- We know that each row of the linkage matrix merges two branches into one. We replace clusters in  $(y_{m,1}, y_{m,2})$  with their constituents recursively, until no clusters remain.
- These replacements preserve the order of the clustering.
- The output is a sorted list of original(unclustered) items.



[Fuente imagen](#)

# HIERARCHICAL RISK PARITY

3 pasos

- **recursive bisection**

Hace uso de la quasidiagonalización bottom-up del paso anterior

porque hace inverse variance weightings que son óptimos para una matriz diagonal!

Los pesos suman 1 y siempre están entre 0 y 1 porque vamos dividiendo recursivamente los pesos recibidos desde un nivel jerárquicamente superior

Los splits en los weights son en inverse proportion a la varianza del cluster

Fuente: Marcos López de Prado

Advances in Financial Machine Learning

1. The algorithm is initialized by:

(a) setting the list of items:  $L = \{L_0\}$ , with  $L_0 = \{n\}_{n=1,\dots,N}$

(b) assigning a unit weight to all items:  $w_n = 1, \forall n = 1, \dots, N$

2. If  $|L_i| = 1, \forall L_i \in L$ , then stop.

3. For each  $L_i \in L$  such that  $|L_i| > 1$ :

(a) bisect  $L_i$  into two subsets,  $L_i^{(1)} \cup L_i^{(2)} = L_i$ , where  $|L_i^{(1)}| = \text{int}[\frac{1}{2}|L_i|]$ , and the order is preserved

(b) define the variance of  $L_i^{(j)}, j=1,2$ , as the quadratic form  $\tilde{V}_i^{(j)} \equiv \tilde{w}_i^{(j)'} V_i^{(j)} \tilde{w}_i^{(j)}$ , where  $V_i^{(j)}$  is the covariance matrix between the constituents of the  $L_i^{(j)}$  bisection, and  $\tilde{w}_i^{(j)} = \text{diag}[V_i^{(j)}]^{-1} \frac{1}{\text{tr}[\text{diag}[V_i^{(j)}]^{-1}]}$ , where  $\text{diag}[\cdot]$  and  $\text{tr}[\cdot]$  are the diagonal and trace operators

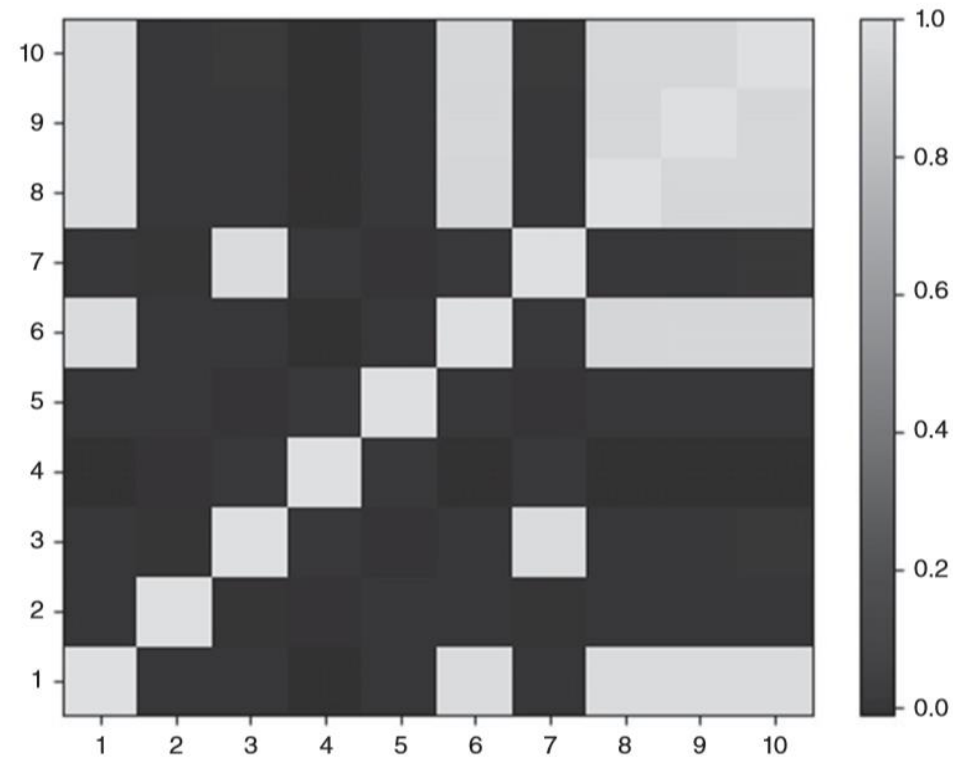
(c) compute the split factor:  $\alpha_i = 1 - \frac{\tilde{V}_i^{(1)}}{\tilde{V}_i^{(1)} + \tilde{V}_i^{(2)}}$ , so that  $0 \leq \alpha_i \leq 1$

(d) re-scale allocations  $w_n$  by a factor of  $\alpha_i, \forall n \in L_i^{(1)}$

(e) re-scale allocations  $w_n$  by a factor of  $(1 - \alpha_i), \forall n \in L_i^{(2)}$

4. Loop to step 2

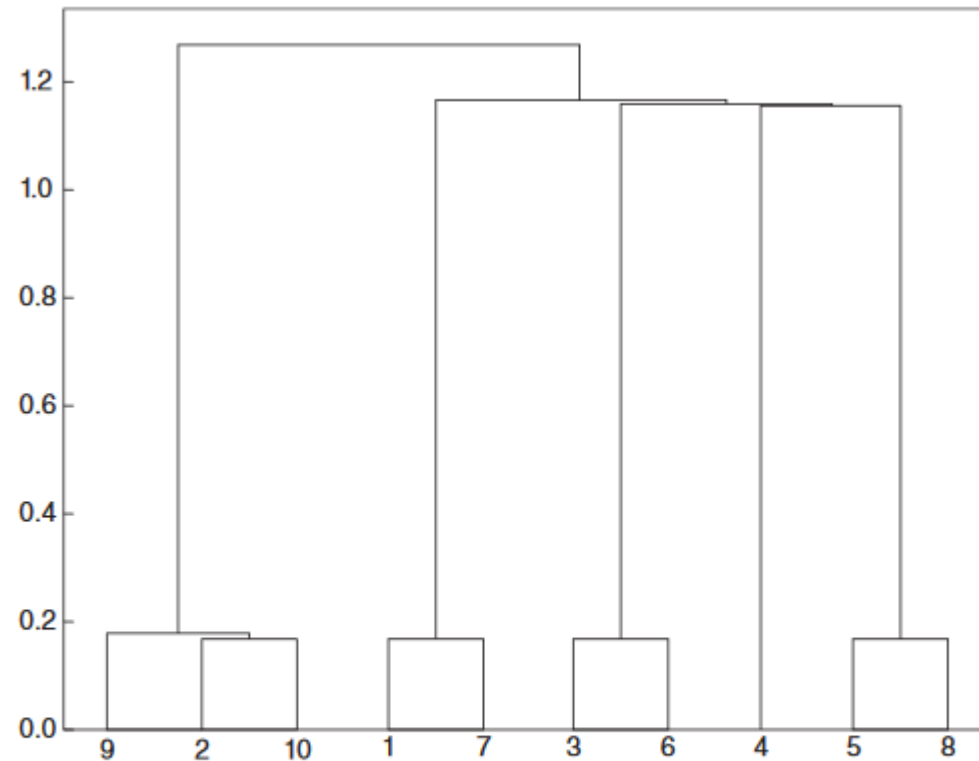
## HIERARCHICAL RISK PARITY



**FIGURE 16.4** Heat-map of original covariance matrix

This correlation matrix has been computed using function `generateData` from snippet 16.4 (see Section 16.A.3). The last five columns are partially correlated to some of the first five series.

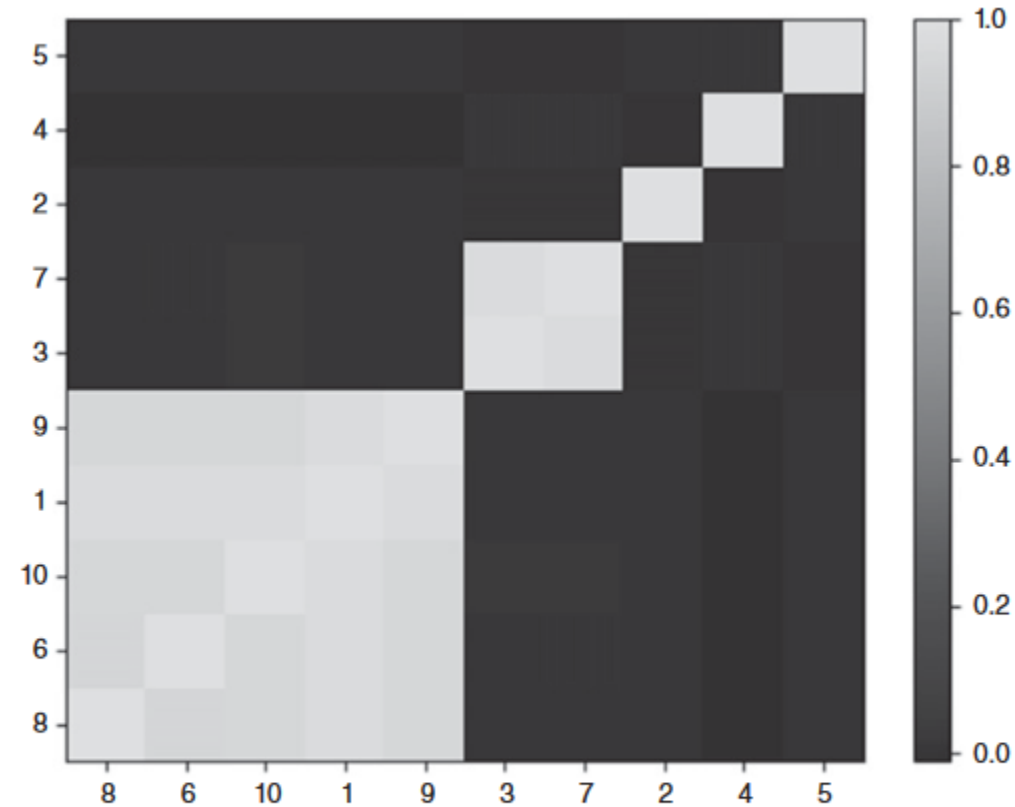
## HIERARCHICAL RISK PARITY



**FIGURE 16.5** Dendrogram of cluster formation

The clustering procedure has correctly identified that series 9 and 10 were perturbations of series 2, hence (9, 2, 10) are clustered together. Similarly, 7 is a perturbation of 1, 6 is a perturbation of 3, and 8 is a perturbation of 5. The only original item that was not perturbed is 4, and that is the one item for which the clustering algorithm found no similarity.

## HIERARCHICAL RISK PARITY



**FIGURE 16.6** Clustered covariance matrix

Stage 2 quasi-diagonalizes the correlation matrix, in the sense that the largest values lie along the diagonal. However, unlike PCA or similar procedures, HRP does not require a change of basis. HRP solves the allocation problem robustly, while working with the original investments.

Fuente: Advances in Financial Machine Learning. Marcos López de Prado.

HIERARCHICAL RISK PARITY. RESULTADOS DE UN EXPERIMENTO DE MONTECARLO.

TABLE 16.1 A Comparison of Three Allocations

|                                          | Weight #                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CLA    | HRP    | IVP    |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|
| En la comparación usa constraints        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 14.44% | 7.00%  | 10.36% |
|                                          | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 19.93% | 7.59%  | 10.28% |
|                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 19.73% | 10.84% | 10.36% |
|                                          | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 19.87% | 19.03% | 10.25% |
|                                          | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 18.68% | 9.72%  | 10.31% |
| $0 \leq w_i \leq 1$ (non-negativity)     | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.00%  | 10.19% | 9.74%  |
|                                          | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5.86%  | 6.62%  | 9.80%  |
|                                          | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.49%  | 9.10%  | 9.65%  |
|                                          | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.00%  | 7.12%  | 9.64%  |
|                                          | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.00%  | 12.79% | 9.61%  |
| $\sum_{i=1}^N w_i = 1$ (full investment) | A characteristic outcome of the three methods studied: CLA concentrates weights on a few investments, hence becoming exposed to idiosyncratic shocks. IVP evenly spreads weights through all investments, ignoring the correlation structure. This makes it vulnerable to systemic shocks. HRP finds a compromise between diversifying across all investments and diversifying across cluster, which makes it more resilient against both types of shocks. |        |        |        |

Fuente: Advances in Financial Machine Learning. Marcos López de Prado.

## HIERARCHICAL RISK PARITY. RESULTADOS DE UN EXPERIMENTO DE MONTECARLO.

CLA concentrates 92.66% of the allocation on the top-5 holdings, while HRP concentrates only 62.57%.

Second, CLA assigns zero weight to 3 investments (without the  $0 \leq w_i$  constraint, the allocation would have been negative).

HRP seems to find a compromise between CLA's concentrated solution and traditional risk parity's IVP allocation.

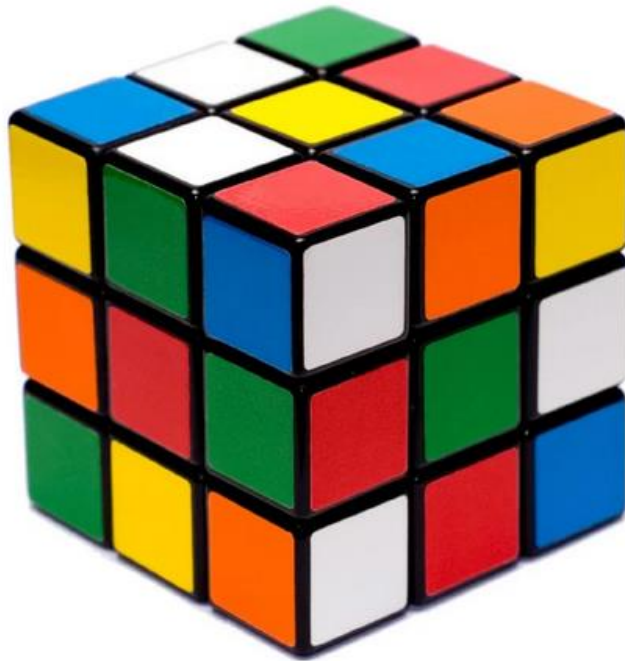
.What drives CLA's extreme concentration is its goal of minimizing the portfolio's risk.

And yet both portfolios have a very similar standard deviation ( $\sigma_{HRP}=0.4640, \sigma_{CLA}=0.4486$ ).

So CLA has discarded half of the investment universe in favor of a minor risk reduction. The reality of course is that CLA's portfolio is deceitfully diversified, because any distress situation affecting the top-5 allocations will have a much greater negative impact on CLA's than on HRP's portfolio.

Fuente: *Advances in Financial Machine Learning*. Marcos López de Prado.

## OTRO PROBLEMA...



¿Será fácil estimar las matrices de covarianzas?

Claro que no!

Ver Cap 2 de Machine learning for asset managers de MLP

## ASSET ALLOCATION

- Siempre queda algo más...
- En los últimos años se han refinando las técnicas para el denoising y detoning, filtrado del componente de mercado, de la matriz de covarianzas (López de Prado 2020)
- El uso de una matriz de covarianzas de los retornos para la optimización con ruido da como resultado weights óptimos imprecisos mientras que la existencia de un componente común dificulta el proceso de clustering.

